Ex. No. : 15	Air Pollution Monitoring System
Date: 13-10-2025	

Aim

To implement an Air Pollution Monitoring System using ESP32

Algorithm

Step 1: Initialize ESP32 pins:
  i. Potentiometer (simulated gas sensor) → Analog input (A0).
  ii. LED (optional) → Output to indicate pollution alert.

Step 2: Read analog value from the sensor.

Step 3: Compare with threshold:
  i. If air quality value is high → mark status "Polluted".
  ii. Else → mark status "Good".

Step 4: Display readings on Serial Monitor.

Step 5: Repeat continuously.

Source Code

Include Library: MQ2_LPG

```
#define GAS_SENSOR_PIN 34   // Analog input pin for MQ135 simulation

#define ALERT_LED 2         // Built-in LED for pollution alert


int threshold = 2000; // Adjust threshold (0–4095 on ESP32 ADC)


void setup() {
  Serial.begin(115200);
  pinMode(ALERT_LED, OUTPUT);
}


void loop() {
  int airValue = analogRead(GAS_SENSOR_PIN); // Read air quality value
  Serial.print("Air Quality Value: ");
  Serial.println(airValue);
```

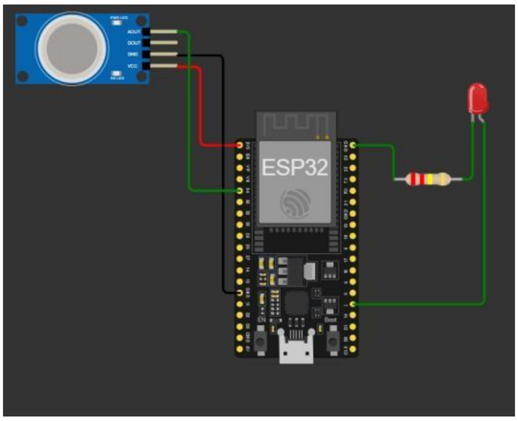
```
  if (airValue > threshold) {
    digitalWrite(ALERT_LED, HIGH);
    Serial.println("Status: Polluted Air");
  } else {
    digitalWrite(ALERT_LED, LOW);
    Serial.println("Status: Good Air");
  }

  Serial.println("------------------------------");
  delay(2000); // Check every 2 seconds
}
```

Circuit Connection

Component	ESP32 Pin	Description
MQ135 VCC	3.3V	Power supply for gas sensor
MQ135 GND	GND	Common ground
MQ135 AOUT (Analog Out)	GPIO 34	Reads analog gas level
LED (Anode +)	GPIO 2	Indicates pollution alert
LED (Cathode -)	GND (through 220Ω resistor)	Current limiting path

Device Setup



Output

Air Quality Value: 1500

Status: GoodAir

- - - - - - - - - - - - - - - - - - -

Air Quality Value: 3200

Status: Polluted Air

- - - - - - - - - - - - - - - - - - -

Result:

Thus the program has been executed and output verified successfully.